

Model Selection — Complete Guide

From Problem to Model — Step by Step

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Step 0: From Problem to Data

You don't need data to CHOOSE the model. You need the PROBLEM.

The problem tells you the model. The model tells you what data to collect.

The Flow

*PROBLEM → ANSWER TYPE → DATA NEEDED → SOURCE → AMOUNT
→ MODEL*

Answer Type → Data to Collect

Answer Type	Data Needed	Labels?
WORD (spam/yes/no)	Labeled examples	YES
NUMBER (Rs 72 lakhs)	Past records with numbers	YES
GROUPS (customer types)	Just raw data	NO
WEIRD (anomaly)	Normal examples only	NO
STRATEGY (RL)	Simulator / environment	NO

How Much Data?

Model	Minimum	Ideal
Logistic Reg / DT	500 rows	5K+ rows
Random Forest	1K rows	10K+ rows
CNN from scratch	10K images	100K+ images
CNN + Transfer Learning	200 images	1K+ images
K-Means / DBSCAN	500 rows	5K+ rows
Autoencoder	1K normal examples	10K+ normal
RL	1M simulated episodes	More = better

Real Walk-Throughs

ExamGuard Phones: Answer: Phone/No Phone = WORD. Need labeled IMAGES. Take 500 photos + augment. Model: YOLO + Transfer Learning.

Predict Sales: Answer: Rs amount = NUMBER. Need past sales records from database. Model: Linear Regression first.

Customer Groups: Answer: Groups = CLUSTERING. Need customer info from CRM. No labels. Model: K-Means (K=3).

Unusual Behavior: Answer: Normal/Unusual but can't label unusual! Need normal exam footage. Model: Autoencoder.

Product Reviews: Answer: Positive/Negative = WORD. Need 20K labeled reviews. Model: Naive Bayes (text data).

Self-Driving: Answer: Strategy over time. Need driving simulator. Define rewards. Model: Reinforcement Learning.

Common Mistakes

- "I'll collect ALL data first" → Decide model type FIRST, then collect the RIGHT data.
- "I need millions of records" → Many models work with 500-5K. Check minimum table.
- "I have data but no problem" → Always start with PROBLEM, not data.
- "I'll use Deep Learning for everything" → DL needs 100K+. On 5K it overfits.

Steps 1-5: The 5 Questions (When You Have Data)

Q1: What does the answer look like?

WORD = Classification | NUMBER = Regression | GROUPS = Clustering | STRATEGY = RL

Q2: Do I have labels?

YES = Supervised | NO = Unsupervised | REWARDS = RL

Q3: What type of data?

IMAGES = CNN/YOLO | TEXT = Naive Bayes/BERT | NUMBERS = Many options

Q4: How much data?

<5K = Simple models | 5K-100K = Most models | 100K+ = Deep Learning

Q5: Special needs?

Explain? = Decision Tree | Real-time? = YOLO | Imbalanced? = Anomaly Detection

12 Cases — Quick Summary

Case 1: Logistic Regression

WORD + Labels + Numbers + Simple/Small + 2 categories → Logistic Regression. Examples: Email spam, employee quit, ad click.

Case 2: Decision Tree

WORD + Labels + Numbers + Must EXPLAIN → Decision Tree. Examples: Diabetes diagnosis, car insurance, loan approval.

Case 3: Random Forest

WORD + Labels + Numbers + Best accuracy (no explain) → Random Forest. Examples: Disease diagnosis, product defects.

Case 4: CNN/YOLO

WORD + Labels + IMAGES → CNN. Real-time? YOLO. <500 images? Transfer Learning. Examples: ExamGuard, traffic.

Case 5: Imbalance Trap

99% one class → DON'T classify! Use Isolation Forest. Examples: Credit card fraud, factory defects.

Case 6: Naive Bayes

WORD + Labels + TEXT → Naive Bayes. Examples: Product reviews, support tickets.

Case 7: Linear Regression

NUMBER + Labels → Always start Linear. If $R^2 > 0.80$, DONE. Examples: House prices, salary.

Case 8: Polynomial Regression

NUMBER + Linear bad + Data CURVED → Polynomial. Examples: Crop yield, ad spend ROI.

Case 9: K-Means vs DBSCAN

No labels + Groups. Know K? K-Means. Don't know? DBSCAN. Examples: Customer segments, GPS routes.

Case 10: Autoencoder

No labels + Find weird + Images → Autoencoder. Examples: ExamGuard unusual, server monitor.

Case 11: RL Strategy

Sequential decisions + affects future + rewards → RL. Examples: Self-driving, warehouse robot.

Case 12: RL Alerts

Balance competing goals + context matters → RL. Examples: ExamGuard alerts, ICU monitor.

6 Expert Tricks

Start Simple: Try the simplest model first. If LogReg gives 83% and CNN gives 85%, use LogReg.

Data Size Matters: 500 patients? Decision Tree. 500K X-rays? CNN. Match model to data size.

Check What Others Use: Google "object detection" → everyone uses YOLO. Don't reinvent.

Transfer Learning: YOLO pre-trained on millions. Fine-tune with your 500 photos. Magic.

Break Into Sub-Problems: ExamGuard = 7 sub-problems. Each needs a different model.

Test Multiple: Try 3-4 models. Compare accuracy + speed + explainability. Pick best balance.